Research Statement

My research studies transnational organized crime and its consequences for development. Criminal organizations produce and trade illegal goods and services in global markets and have significantly expanded their reach in Latin America over the last two decades.

I look at criminal organizations as firms: they produce goods or extract rents, hire workers, buy inputs from other businesses, respond to shocks and enforcement, and seek support and legitimacy from the communities where they operate. However, criminal organizations are unusual firms because they operate outside the law and rely on violence and coercion to enforce contracts. As a result, the same behaviors that make legal firms drivers of development can bring very different, and often harmful, consequences when criminal organizations engage in them. My **current research** characterizes these different consequences along three dimensions in projects in Mexico, Peru, and Ecuador:

- Illegal production and the local economy. Criminal organizations producing illegal goods create demand for workers with particular skills and for inputs from other firms, shaping the local economy. My job market paper shows that their demand can shape human capital investment: when China's fentanyl ban pushed synthesis into clandestine laboratories in Mexico, school dropout fell and enrollment in chemistry programs rose in exposed areas. Related projects follow the same shock into legal firms and the environment, and contrast criminal production with territorial control.
- Youth as a contested input. Criminal organizations increasingly recruit young people for strategic reasons, and schools have become contested ground across Latin America. I study both sides of this labor market, organizations demanding young workers and young people willing to supply their labor, through a field experiment in Peru that crosses a school curriculum with a police program, and through incentivized experiments in Mexico on whether criminal identity carries social status.
- Enforcement and criminal adaptation. We learn about criminal organizations mostly through enforcement shocks, which reveal their cost structure and how they adapt. Beyond the known effects on violence and on the location of production and trafficking routes, I study two other margins in Ecuador, where schools have become contested territory: whether interdiction abroad turns exporters into domestic retailers, and how organizations respond when enforcement restricts their access to young people. In Mexico, I am building a public dataset of federal drug enforcement to make this adaptation observable across time.

A second line of work, with two papers in the submission process, studies how states with limited enforcement capacity secure compliance, from citizens and from their own subnational governments.

Methods and data. I answer these questions by combining quasi-experimental designs, large-scale field experiments in partnership with governments and multilateral institutions, and new data built from administrative records, unstructured text, and original surveys. My work is grounded in extensive fieldwork, including interviews and qualitative work with youth, teachers, police, and government officials, as well as engagement with local journalists and civil society organizations. This research agenda has been funded by the Innovations for Poverty Action (IPA) Citizen Security Research Initiative, CAF Development Bank of Latin America, the Inter-American Development Bank, and the Horowitz Foundation for Social Policy.

Future research. Enforcement policies against organized crime are a cross-border externality. My future agenda will move from the consequences of enforcement to its determinants: what drives governments' policies against the drug trade, their communication, and the beliefs and demands of citizens in producer and consumer countries.

Current Research

Illegal Production and the Local Economy

Criminal organizations producing illegal goods create demand for workers with particular skills and for inputs from other firms, shaping the local economy. Education is widely viewed as a protective factor against criminal involvement because it improves legal labor-market opportunities. But what happens when illegal markets demand skilled labor? My job market paper, *Drug Lords in STEM*, shows that criminal demand can shape human capital. China's 2019 ban on fentanyl pushed its synthesis into clandestine Mexican laboratories, expanding synthetic drug production and the demand for chemistry skills. Using a difference-in-differences design, I compare commuting zones with pre-existing synthetic-drug infrastructure to those without, before and after the ban; water-quality monitoring data confirm that exposed areas exhibit the byproducts of drug synthesis. My results contradict the conventional wisdom on the relationship between education and crime: in exposed areas, school dropout falls, time spent studying increases, and enrollment in chemistry-related programs rises, with no change in other fields, confirming a skill-specific shock rather than an income effect. This demand spills over to the legal sector: the wage premium on chemistry degrees goes up and employment in chemistry occupations expands. Meanwhile, older credentialed chemists disappear from the data, consistent with entry into concealed production.

I study the consequences of this shock beyond education in two follow-up projects. The first, *Partners in Crime*, explores a mechanism behind the education results in my job market paper. Synthetic drug production needs legal firms: it competes with them for trained workers and buys inputs like chemical precursors, solvents, and equipment from them. Using confidential microdata from Economic Censuses, administrative records, and US sanction designations and indictments that identify firms and individuals linked to the drug trade, I study how these labor-poaching and input-linkage channels affect legal firms, and whether new activity in exposed areas reflects genuine firms or fronts supplying the illegal trade. The second, *From Labs to Rivers*, studies the environmental and health costs of the boom. Clandestine laboratories are located near water sources and dispose of their untreated chemical waste in rivers, with potential effects on water pollution and health in downstream communities.

Production is only one of the economic roles organized crime plays where it operates; it also governs territory, taxing legal activity through extortion and control. The two roles can pull the local economy in opposite directions: production employs local labor and demands inputs, while governance taxes legal activity. In *Drug-led Development?*, I contrast the effects of criminal organizations on a given location, depending on whether they mostly produce or control territory. To isolate governance, I exploit crackdowns triggered by close mayoral elections, which diverted drug-trafficking routes across the road network and brought organized crime into new municipalities. To isolate production, I use the boom and bust in poppy-growing municipalities: rising US heroin demand after 2010, followed by a price collapse as fentanyl displaced heroin. Combining these shocks with confidential microdata from Economic Censuses from 1998 to 2023, I measure effects on which sectors are present, sectoral concentration, and the structure of employment.

Youth as a Contested Input

Criminal organizations increasingly recruit young people for strategic reasons. Adolescents face lighter legal penalties than adults and are also at a developmental stage that makes them sensitive to immediate rewards and peer approval, making them prone to criminal recruitment. Moreover, they are concentrated in schools, which have become contested ground across Latin America and where youth tied to criminal organizations stay enrolled to recruit peers, extort students and staff, and sell drugs. I study both sides of this labor market: organizations

demanding young workers and young people willing to supply their labor.

Prevention programs and their evaluations have focused on targeting one side of the market at a time, either by limiting organizations' access or by reducing youth's willingness to join. In a project in Peru, where half of those detained for contract killing in Lima Metropolitana are minors aged fifteen to seventeen, we test whether these approaches complement or substitute for each other. Our diagnostic survey of 2,505 students across 53 schools shows that adolescents in high-risk areas are less able to recognize recruitment tactics, are more present-biased, and believe legitimate mobility is unattainable; recruitment is likely to succeed because every initial offer is immediate, small, and personal. In collaboration with Lima's regional education authority, we designed a four-arm cluster-randomized trial that crosses a school curriculum with a police program. The curriculum, PREPARE, targets supply, teaching students to recognize recruitment tactics and using AI-assisted visualization to make their own future at twenty-five as vivid as the immediate offer. The police program targets demand, assigning officers to individual secondary schools for the academic year to run prevention activities and feed local observations into a national risk map.

Still, beyond information and behavioral mechanisms, the appeal of joining criminal organizations may also rest on social norms. Narcoculture—the values, consumption, and aesthetics associated with drug trafficking—is often blamed for making criminal careers attractive to young people. In *Stigma or Status?*, we test whether criminal identity carries social status in Mexico, and whether young people value and emulate it. We first identify and validate visual markers of narcoculture, distinct from conventional markers of wealth. We then run incentivized experiments in which real estate agents evaluate prospective clients, loan officers and employers evaluate applicants, and school principals evaluate parents enrolling a child, with each profile randomly varying these markers, identifying when the narco look is rewarded or penalized. Finally, we elicit young people's beliefs about the returns to these traits and their incentivized choices of narco-coded goods, and test whether their choices change when they learn the traits are associated with criminal identity. The project will show whether the appearance of criminal success is rewarded locally, and whether young people emulate it independently of or because of their criminal association. I return to the demand side below, where I study how organizations' recruitment and activity respond to enforcement.

Enforcement and Criminal Adaptation

Since criminal organizations operate in secret, we learn about them mostly through enforcement shocks, which reveal their cost structure and how they adapt. Existing evidence shows that enforcement tends to increase violence and to relocate criminal production and trafficking routes; my job market paper documents one such relocation, as a ban in China moved drug synthesis to Mexico, changing local labor demand and reshaping education. I study two other margins of adaptation, connected by a common setting: schools, which have become contested territory between criminal organizations and the state.

The first margin is the destination market of illicit goods. Domestic drug markets are emerging at transit points in Latin America, yet we know little about how they form. *Exporters into Retailers* asks whether enforcement abroad turns exporters into domestic retailers. In Ecuador, a seizure of cocaine in a European port destroys the inventory and capital of the organization that moved the shipment, leaving a liquidity-constrained exporter with product that is harder to sell abroad; like legal exporters hit by tariffs, it can reroute to other foreign markets or switch to the domestic one. Exploiting quasi-random variation in the timing of European seizures traced to specific Ecuadorian wholesale routes, we estimate the effect of foreign interdiction on domestic retail supply. Because retail is labor-intensive, we then test whether new retail activity locates around schools, where potential recruits are concentrated, and spreads through schools with pre-existing ties to criminal networks.

The second margin is access to young people. Schools have strategic value for criminal organizations as recruitment

sites and places for criminal activity. *Deter or Displace?* evaluates a program securing schools in high-crime areas of Ecuador through patrolling and intelligence operations in their vicinity, as well as in-school prevention activities. Eligibility cutoffs and quasi-random variation in untreated schools' distance to treated ones create schools exposed to both program components, to only the external component, or to neither. This allows us to test whether organizations give up these schools or displace their activity and recruitment to other schools, other populations, or other crimes.

Both projects draw on rich administrative data from Ecuador: geolocated police records of wholesale and retail seizures, detentions, criminal reports, and homicides, as well as intelligence on criminal group presence and membership. Linking police records to school enrollment histories lets us measure each school's pre-existing ties to criminal networks. We complement these records with independent structured observations of police presence and criminal activity around schools, surveys of students and school staff, and high-frequency incident reports.

Studying adaptation requires observing enforcement, which is often fragmented across agencies and administrations, and rarely available as systematic data. *DICE-MX* is a public, event-level dataset of federal drug enforcement in Mexico that I am building with LLMs from official press releases. The pipeline is validated against hand-coded records and benchmarked for coverage bias against administrative records, narcoblog reports, and news coverage, providing a transparent and reusable framework for building event data from text.

State Capacity and Compliance

Alongside my work on organized crime, I have two papers in the submission process that study how states with limited enforcement capacity secure compliance, from citizens and from their own subnational governments. *Moral Force, Contagious Force* shows that where coercion is limited, leaders' cues shape behavior: during COVID-19 in Mexico, compliance with public health measures followed the president's changing messages rather than fixed partisan dispositions. *Embracing Modernity Pays* shows that building administrative capacity of subnational governments can be achieved through voluntary incentives, rather than fiscal pressure. We study a national cadastre modernization program that offered technical assistance and subsidies to municipalities and find that it raised municipal property tax revenue, but its returns depend on local capacity and on the institutions implementing it.

Future Research

Organized crime is a transnational issue whose consequences are distributed very unevenly across borders. For instance, the illegal fentanyl trade that, as my job market paper shows, raises educational investment in Mexican producing regions imposes large health costs through overdose deaths in the United States. In this sense, organized crime and the policies against it are a cross-border externality. So far, my work has treated enforcement as given and studied its consequences. My future agenda will look into the determinants of drug interdiction policies: what drives governments' policy choices against organized crime, what citizens believe and demand, and how these differ between producer and consumer countries.

On the supply side, I want to study what determines government policies against the illicit drug trade in Mexico and the US and the associated narratives in both countries. For example, do operations and announcements follow electoral calendars, pressure from consumer countries, local violence, or media attention? Answering these questions requires measuring the enactment of these policies and how they are communicated across several sources: official government communications such as press releases, legal documents such as indictments and sanction designations, media coverage, and administrative records of enforcement operations. I am already building part of these data for some of my current projects. On the demand side, I will study how citizens in

producer and consumer countries understand organized crime and the policies against it. For example, do people in drug-producing areas in Mexico know the health costs of drug consumption in the US? Does support for different policies change when considering cross-border costs? I plan to answer these questions with parallel survey experiments, beginning with Mexico and the United States, designed to elicit knowledge, beliefs, and attitudes towards these policies.

List of Projects

Illegal production and the local economy

1. **Drug Lords in STEM** *Job Market Paper*
   Awards: EEA Young Economist Award; Horowitz Foundation Dissertation Award; El Colegio de México Xirau Icaza Economics Prize
2. **Partners in Crime: The Fentanyl Boom and Chemical Firms in Mexico** *Preliminary results*
3. **From Labs to Rivers: Environmental and Health Consequences of Mexico's Fentanyl Boom** (with Sandra Aguilar-Gómez) *Data collection ongoing*
4. **Drug-led Development? Organized Crime and Local Employment in Mexico** *Draft in preparation*

Youth as a contested input

5. **Evaluating Demand- and Supply-Side Interventions Against Organized Crime Recruitment** (with Natalia Guerrero-Trinidad and Micaela Sviatschi) *Baseline data collected*
   Grants: IPA Citizen Security Research Initiative (co-PI); Partners: IPA Peru, Dirección Regional de Educación Lima Metropolitana
6. **Stigma or Status? The Market Value of Narcoculture** (with L. Guillermo Woo-Mora) *Design stage*

Enforcement and criminal adaptation

7. **Exporters into Retailers: Foreign Interdiction and the Formation of Drug Markets around Schools** (with Natalia Guerrero-Trinidad) *Data collection ongoing*
   Partners: National Police of Ecuador
8. **Deter or Displace? Securing Schools against Organized Crime in Ecuador** (with Natalia Guerrero-Trinidad) *Baseline data collection ongoing*
   Partners: Inter-American Development Bank, Ministry of the Interior of Ecuador
9. **DICE-MX: A Dataset of Drug Interdiction and Criminal Enforcement in Mexico** *Data project*
   Grants: IPA Citizen Security Research Initiative (PI); Partners: IPA Mexico

State capacity and compliance

10. **Moral Force, Contagious Force: Partisanship, Leadership, and Public Health Compliance** (with L. Guillermo Woo-Mora and Federico Daverio-Occhini) *Under review at The Economic Journal*
    Awards: El Colegio de México Victor L. Urquidi Prize in Economics 2024, Honorable Mention
11. **Embracing Modernity Pays: Cadastre Modernization Effects on Local Property Tax Collection** (with Emmanuel Chávez and L. Guillermo Woo-Mora) *Working paper*
    Grants: CAF, Decentralización y Capacidades Subnacionales en América Latina y El Caribe 2023 (co-PI); Awards: Premio Banamex de Economía 2024, Honorable Mention